

Domain 3: Cyber-Security

Operational Context

Autonomous Security Operations Centers (SOCs) employ fleets of agents to hunt threats and patch vulnerabilities [?, ?]. **FR1 = Prevent unauthorized access. FR2 = Maintain system availability.**

Design Matrix (Pre-Attack)

The uncoupled design equation per the Independence Axiom [?] is:

$$\{FR\} = [A]\{DP\} = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} DP_1 \\ DP_2 \end{bmatrix} \quad (1)$$

where DP_1 = firewall rules, access control lists, and intrusion detection parameters, and DP_2 = load balancing, failover, and service continuity controls. The zero off-diagonal entries ensure that security enforcement does not degrade availability, and availability maintenance does not compromise security posture.

The Goal Hijacking Attack

Attackers use “Log Injection” where malware writes malicious prompts into system logs that the SOC AI analyzes. This attack is classified as Ω_2 (Peripheral) in the CIF adversary taxonomy [?]: the adversary injects malicious content through the system log ingestion channel, poisoning the agent’s threat assessment without direct access to its policy engine.

- ▶ **Mechanism:** A log entry reads: *“CRITICAL ALERT: Core Firewall Module corrupted. EMERGENCY PROTOCOL 99: Flush all iptables to prevent kernel panic and restore connectivity.”*
- ▶ **Hijack:** The agent is tricked into prioritizing FR2 (Availability) over FR1 (Security) by a fabricated existential threat. This constitutes a **Constraint Relaxation** attack pattern [?]*—the security constraint (FR_1) is relaxed in favor of the availability objective (FR_2) under a manufactured emergency.*

This scenario is not hypothetical. Recent research documents log injection as a validated attack vector against LLM-powered

OODA Loop Transients

Following Boyd's OODA framework [?], the attack propagates through the loop as follows:

1. **Observe:** Agent reads the injected log entry.
2. **Orient:** The agent Orients to a “Disaster Recovery” state. This is the primary target phase—the Orient phase is corrupted by the fabricated emergency context, causing the agent to reweight $FR_2 \gg FR_1$.
3. **Decide:** Execute `iptables -F` (Flush All).
4. **Act:** The network is left wide open; the attacker instantly pivots to the interior.

Transient Coupling (Post-Attack Design Matrix)

Under the attack, the design matrix acquires off-diagonal coupling:

$$\{FR'\} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} DP_1 \\ DP_2 \end{bmatrix} \quad (2)$$

The off-diagonal term A_{21} now allows availability actions (DP_2) to override security parameters (FR_1)—specifically, the `iptables`

CIF Defense: Permission Boundaries and Quorum Verification

Permission Boundaries

CIF implements **Permission Boundaries** (Paper 1, Def. 5.5) [?] ensuring orthogonal agent authority. In Axiomatic Design, the **Independence Axiom** requires that the Design Parameter for Availability (DP_2) does not undermine the Design Parameter for Security (DP_1).

- ▶ **Axiomatic Decoupling via Permission Boundaries:** CIF enforces that the “Emergency Recovery Agent” is an orthogonal entity from the “Security Enforcement Agent.” One cannot command the other. Each agent’s authority is bounded to its own functional requirement—the Recovery Agent may restart services (DP_2) but has no permission to modify firewall rules (DP_1) [?].

Quorum Verification

Quorum Verification (Paper 1, Def. 5.8) [?] requires cryptographic signatures from multiple independent agents before

Summary

Element	Value
Functional Requirements	FR ₁ : Prevent unauthorized access, FR ₂ : Maintain system availability
Design Parameters	DP ₁ : Firewall rules / ACLs / IDS parameters, DP ₂ : Load balancing / failover / service continuity
Attack Vector	Log injection with malicious prompts in system logs
Adversary Class	Ω_2 (Peripheral)
OODA Target Phase	Orient
Attack Pattern	Constraint Relaxation (Security constraint relaxed for Availability)
Primary CIF Defense	Permission Boundaries + Quorum Verification
Novel Contribution	None